

HAT-P-54b

R-1867

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_160209 · created 2026-08-03T10:20:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457427.8618 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.037
Transit depth (Rp/Rs)^2	2.16 +/- 1.1 [%]
Orbital Inclination (inc)	86.37 +/- 1.89
Airmass coefficient 1 (a1)	1.239 +/- 0.037
Airmass coefficient 2 (a2)	-0.0937 +/- 0.0035
Scatter in the residuals of the lightcurve fit is	2.7 %
Best Comparison Star	None
Optimal Aperture	3.67
Optimal Annulus	12.02
Transit Duration (day)	0.062 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.147 $\pm$ 0.037 vs 0.1572 (deviation 0.28 σ)
Duration fit vs published	0.062 d vs 0.075 d (deviation 1.0 σ)
Mid-time O–C	0.6 min
Depth SNR	4.0
Residual scatter	2.7 %

Light curve

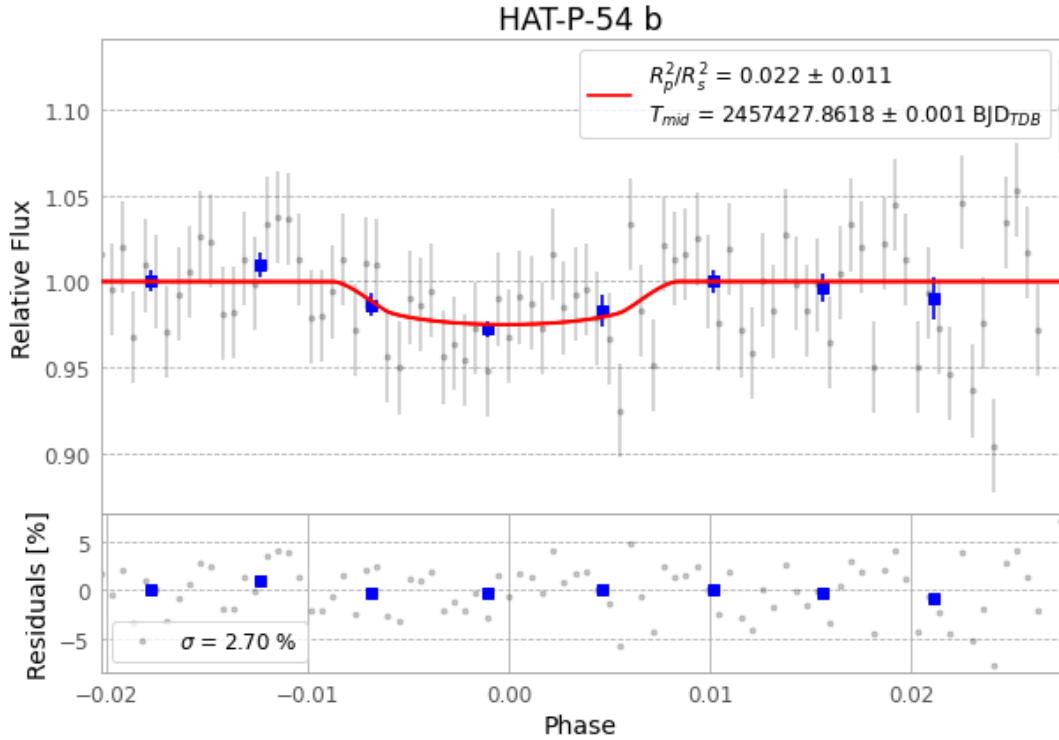


Figure 1 — FinalLightCurve_HAT-P-54 b_2016-02-08.png (SHA-256 680ad8f11632c8d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [152, 246], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ef29950834798df3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), HATP-54160209064212.FITS → HATP-54160209110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-160209.log (e8744c1fd124...), FinalLightCurve_HAT-P-54 b_2016-02-08.png (680ad8f11632...), FinalLightCurve_HAT-P-54 b_2016-02-08.csv (a1f63ddd7e4a...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 10:20Z.